

available at www.sciencedirect.com
journal homepage: www.europeanurology.com



Fast Track Submission*

Safety and Efficacy of Tranexamic Acid in Urologic Surgery: Results from the International, Randomized, Placebo-Controlled POISE-3 Trial

Kari A.O. Tikkinen^{a,b,c,*}, Maura Marcucci^{c,d,e}, Alex L.E. Halme^b, Sandra N. Ofori^{e,f},
Flavia K. Borges^{c,e,f,g}, Charlotte T.B. Kanstrup^h, Lily J. Park^{e,i}, Farhad Tondroanagam^b,
Tuomas P. Kilpeläinen^a, Sara V. Tornberg^a, Thomas W. Painter^j, David Conen^{e,f,g},
Vladimir Lomivorotov^{k,l}, Daniel I. Sessler^m, Matthew T.V. Chanⁿ, Maria Jose Martinez-Zapata^o,
Chew Y. Wang^p, Denis B. Xavier^{q,r}, Michael K. Wang^{c,e,f}, Wojciech Szczeklik^s, Kate Leslie^t,
Luke T. Lavalley^u, Rodney H. Breau^u, Kumar Balasubramanian^e, P.J. Devereaux^{c,e,f,v}

^a Department of Urology, University of Helsinki and Helsinki University Hospital, Helsinki, Finland; ^b Faculty of Medicine, University of Helsinki, Helsinki, Finland; ^c Department of Health Research Methods, Evidence and Impact, McMaster University, Hamilton, ON, Canada; ^d Department of Clinical Epidemiology and Research Centre (CERC), Humanitas University & IRCCS Humanitas Research Hospital, Pieve Emanuele, Milan, Italy; ^e Population Health Research Institute, Hamilton, ON, Canada; ^f World Health Research, Hamilton, ON, Canada; ^g Department of Medicine, McMaster University, Hamilton, ON, Canada; ^h Department of Surgery, Copenhagen University Hospital, North Zealand, Hillerød, Denmark; ⁱ Department of Surgery, Division of General Surgery, McMaster University, Hamilton, ON, Canada; ^j University of Adelaide, Adelaide, South Australia, Australia; ^k Department of Anesthesiology and Intensive Care, E. Meshalkin National Medical Research Center, Novosibirsk, Russia; ^l Department of Anesthesiology and Perioperative Medicine, Penn State Milton S. Hershey Medical Center, Penn State College of Medicine, Hershey, PA, USA; ^m Center for Outcomes Research and Department of Anesthesiology, UTHealth, Houston, TX, USA; ⁿ The Chinese University of Hong Kong, Hong Kong Special Administration Region, China; ^o Iberoamerican Cochrane Center, Institut de Recerca Sant Pau, CIBER Epidemiología y Salud Pública, Barcelona, Spain; ^p Department of Anesthesiology, Faculty of Medicine, Universiti Malaya, Kuala Lumpur, Malaysia; ^q Department of Pharmacology, St. John's Medical College, Bangalore, India; ^r Division of Clinical Research and Training, St. John's Research Institute, Bangalore, India; ^s Centre for Intensive Care and Perioperative Medicine, Jagiellonian University Medical College, Krakow, Poland; ^t Department of Critical Care, Melbourne Medical School, University of Melbourne, Melbourne, VIC, Australia; ^u Division of Urology, Department of Surgery, University of Ottawa and Ottawa Hospital Research Institute, ON, Canada; ^v Department of Medicine, Division of Cardiology, McMaster University, Hamilton, ON, Canada

Article info

Article history:

Accepted March 24, 2026

Associate Editor:

Mikkel Fode

Statistical Editor:

Rod Dunn

Abstract

Background and objective: Perioperative bleeding is common. Evidence for tranexamic acid (TXA) in urology remains limited and conflicting, and major urologic guidelines provide no recommendations. In an a priori planned analysis, we evaluated TXA among patients undergoing urologic surgery in POISE-3.

Methods: POISE-3 was an international randomized trial of patients undergoing surgery with bleeding and cardiovascular risk factors. Participants received TXA (1 g intravenous bolus at surgery start and end) or placebo. The primary efficacy outcome was a 30-day bleeding composite (life-threatening/major/critical organ), and the primary safety outcome was a 30-day thrombosis composite (myocardial injury after noncardiac surgery [MINS], nonhemorrhagic stroke, peripheral arterial thrombosis, or symptomatic proximal venous thromboembolism [VTE]).

* Corresponding author. University of Helsinki, Biomedicum 2B, P.O. Box 13, Tukholmankatu 8 B, 00290 Helsinki, Finland.

E-mail address: kari.tikkinen@helsinki.fi (K.A.O. Tikkinen).

<https://doi.org/10.1016/j.eururo.2026.03.019>

0302-2838/© 2026 The Author(s). Published by Elsevier B.V. on behalf of European Association of Urology. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Please cite this article as: K.A.O. Tikkinen, M. Marcucci, A.L.E. Halme et al., Safety and Efficacy of Tranexamic Acid in Urologic Surgery: Results from the International, Randomized, Placebo-Controlled POISE-3 Trial, Eur Urol (2026), <https://doi.org/10.1016/j.eururo.2026.03.019>

Keywords:

Antifibrinolytic agents
 Blood loss
 Blood transfusion
 Cystectomy
 Nephrectomy
 Nephrolithotomy, Percutaneous
 Prostatectomy
 Stroke
 Transurethral resection of prostate
 Thrombosis

Key findings and limitations: Among 1124 urologic participants (556 TXA and 568 placebo), 489 had laparoscopic/robotic, 360 open, 244 transurethral, and 31 percutaneous surgery. The composite bleeding outcome occurred in 45 (8.1%) with TXA and in 62 (10.9%) with placebo (hazard ratio [HR] = 0.73, 95% confidence interval [CI] = 0.50–1.07). Major bleeding occurred in 34 (6.1%) with TXA and in 54 (9.5%) with placebo (HR = 0.63, 95% CI = 0.41–0.97). The composite safety outcome occurred in 67 (12.1%) with TXA and in 62 (10.9%) with placebo (HR = 1.12, 95% CI = 0.79–1.58; MINS: 62 vs 58; strokes: 2 vs 2; VTEs: 3 vs 3). We identified no interactions by surgical approach, cancer status, or recent antithrombotic usage. Patient-important thrombotic events were few, limiting precision for stroke and VTE estimates.

Conclusions and clinical implications: TXA reduced major bleeding and supports perioperative use in urologic surgery, especially when the bleeding risk is high.

© 2026 The Author(s). Published by Elsevier B.V. on behalf of European Association of Urology. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

ADVANCING PRACTICE**What does this study add?**

This a priori planned analysis of urology patients in the international POISE-3 randomized trial provides the largest and most rigorous evidence regarding the efficacy and safety of administering tranexamic acid (TXA) during urologic surgeries. In 1124 patients, the primary composite (bleeding) efficacy outcome occurred in 45/556 (8.1%) in the TXA group and 62/568 (10.9%) in the placebo group (hazard ratio [HR] = 0.73, 95% confidence interval [CI] = 0.50–1.07). TXA reduced major bleeding (34/556 [6.1%] vs 54/568 [9.5%]; HR = 0.63, 95% CI = 0.41–0.97). The primary composite (thrombosis) safety outcome occurred in 67/556 (12.1%) with TXA and 62/568 (10.9%) with placebo (hazard ratio [HR] = 1.12, 95% CI = 0.79–1.58). Stroke and venous thromboembolism (VTE) were uncommon and similar between groups (stroke 2 vs 2; VTE 3 vs 3).

Clinical Relevance

Surveys suggest that TXA remains infrequently used in urologic practice. Our findings have important implications for practice and policy. TXA is inexpensive and easy to administer. Reducing major bleeding can provide meaningful benefits for patients. Given ongoing constraints in blood product availability, wider adoption of TXA in urologic surgery could yield meaningful public health and health system benefits. Estimates for safety were imprecise, with wide confidence intervals that included both important benefit and important harm. However, the totality of perioperative evidence, including nonurologic surgeries, suggests there is no increase in thrombotic risk when using TXA. Guideline panels should consider recommendations on perioperative TXA use in urology, particularly for patients at increased bleeding risk. Associate Editor: Mikkel Fode.

Patient Summary

We tested whether TXA could reduce bleeding complications in patients undergoing urologic surgery. The TXA group had a lower rate of major bleeding compared with the placebo group, while serious safety events were uncommon and occurred at similar rates in the TXA and placebo groups.

1. Introduction

Perioperative bleeding is common in urologic surgery and can lead to major complications, blood transfusions, re-interventions, and prolonged hospitalizations [1–3]. Tranexamic acid (TXA), an antifibrinolytic that inhibits plasmin-mediated fibrin degradation [4], is widely used in orthopedic surgery to reduce perioperative bleeding and transfusion requirements [5–7].

In urology, guidance on TXA remains limited. European Association of Urology and American Urological Association guidelines offer no recommendations for TXA use, and clinicians have not routinely adopted TXA in urologic practice [6–9]. This likely reflects the limited and inconsistent evi-

dence from randomized trials on TXA in urology. In benign urologic surgery, trials in transurethral resection of the prostate (TURP) [10] and percutaneous nephrolithotomy (PCNL) [11] suggest that TXA may reduce transfusions. In oncologic urologic surgery, a placebo-controlled trial in open radical prostatectomy [12] reported lower transfusion rates with TXA, whereas a more recent multicenter Canadian trial in open radical cystectomy found no reduction in transfusion or blood loss [13]. Overall, prior urologic trials had methodological limitations and limited power, particularly for thromboembolic safety outcomes [10,11,14].

POISE-3 was a large, international, placebo-controlled randomized trial comparing intraoperative TXA with placebo in patients undergoing noncardiac surgery at increased

risk of bleeding and cardiovascular events [15]. In the overall study population, TXA reduced a 30-day composite of clinically important bleeding events. However, applicability to urologic surgeries remains uncertain. POISE-3 enrolled more than 1100 patients undergoing urologic surgery, representing the largest randomized evidence base to date for TXA in urology. We therefore evaluated, using an a priori defined analysis plan, the effects of intraoperative TXA versus placebo on 30-day bleeding and thrombosis outcomes among urologic patients.

2. Patients and methods

2.1. Study design

POISE-3 was an international, multicenter, randomized, placebo-controlled trial comparing the efficacy and safety of prophylactic TXA versus placebo in adult patients undergoing noncardiac surgery. Participants, health care providers, including treating clinicians, data collectors, outcome assessors, and adjudicators, were blinded to treatment allocation. Ethics approval was obtained at each participating center before enrollment. The trial was conducted in accordance with the Declaration of Helsinki and Good Clinical Practice and was registered at ClinicalTrials.gov (NCT03505723). This report follows the CONSORT reporting guideline. Details of the trial design have been published [16]. We finalized the statistical analysis plan (Supplement 1) for this a priori planned urologic surgery analysis before any analyses.

2.2. Patients

POISE-3 included participants aged 45 yr or older who underwent inpatient noncardiac surgery, were at increased risk of bleeding and cardiovascular complications according to prespecified criteria (Supplement 2), and provided informed consent [15]. Patients were recruited from June 2018 through July 2021 at 114 hospitals in 22 countries. Exclusion criteria included planned use of systemic TXA during surgery; hypersensitivity or known allergy to TXA; creatinine clearance <30 mL/min or receipt of chronic dialysis; history of seizure disorder; and recent (<3 mo) stroke, myocardial infarction, acute arterial thrombosis, or venous thromboembolism (VTE). Additional exclusion criteria included subarachnoid hemorrhage within the past 30 days; women of childbearing potential who were not taking effective contraception, were pregnant, or breastfeeding.

All POISE-3 participants who underwent urologic surgery were eligible for inclusion in this urology substudy. POISE-3 originally classified all procedures as urologic, general, orthopedic, vascular, spinal, gynecologic, thoracic, plastic, or low-risk surgery [15]. To confirm the specific urologic procedures and surgical approaches, we involved all participating sites that contributed participants undergoing surgery reported to be urologic, general, or low-risk, according to the original POISE-3 definitions, to perform a medical chart review using a prespecified data collection template. We classified the procedure type (for example, radical prostatectomy, pyeloplasty, TURP, PCNL) and the surgical approach (open, laparoscopic/robotic, transurethral, percutaneous) (Supplement 1). We included participants who

underwent procedures that this chart review confirmed or reclassified as urologic and for whom both procedure type and approach could be classified.

2.3. Procedures and data collection

After obtaining informed consent, study personnel randomized participants in a 1:1 ratio to receive a 1 g intravenous bolus of TXA or placebo at the start and end of surgery via an Interactive Web Randomization System (maintained by the POISE-3 Project Office at the Population Health Research Institute in Hamilton, ON, Canada). The randomization process stratified participants by center and applied block randomization with variable block sizes concealed from study personnel. Research personnel followed patients throughout their time in the hospital, evaluating the patients, reviewing their medical records, and recording any outcomes. Study personnel contacted all patients by telephone 30 days after randomization, collected data, and submitted case report forms and supporting event documentation directly into the data management system (Supplement 3).

2.4. Outcomes

Outcomes were assessed at 30 days after randomization. The primary efficacy bleeding (composite) outcome was time to the first event among life-threatening bleeding, major bleeding, or bleeding into a critical organ. The primary safety thrombosis (composite) outcome was time to the first event among myocardial injury after noncardiac surgery (MINS), nonhemorrhagic stroke, peripheral arterial thrombosis, or symptomatic proximal VTE. For this urology surgery analysis, the secondary outcomes included i) bleeding independently associated with mortality after noncardiac surgery (BIMS; defined as bleeding leading to a postoperative hemoglobin <70 g/l, requiring transfusion of red blood cells, and/or judged to be the immediate cause of death), ii) receipt of any transfusion (proportion of patients), and iii) transfusion of ≥ 2 units of packed red blood cells. Tertiary outcomes included i–vi) each component of the primary composites (reported alongside the primary composite outcomes), vii) International Society on Thrombosis and Haemostasis (ISTH) major bleeding, viii) hemorrhagic stroke, ix) myocardial infarction, x) a composite of peripheral arterial thrombosis and amputation, xi) symptomatic pulmonary embolism, xii) symptomatic proximal deep vein thrombosis, xiii) seizures, xiv) rehospitalization for any reason, and xv) death from any cause. Supplement 1 provides detailed definitions. An independent adjudication committee reviewed reported bleeding and thrombotic events.

2.5. Statistical analysis

We finalized the statistical analysis plan on 22 October 2025 before we undertook any analyses (Supplement 1). We analyzed all randomized patients in the groups to which they were assigned, regardless of treatment received or duration of follow-up (intention-to-treat). For all time-to-event outcomes, participants who did not complete 30-day follow-up were censored at the time they left the trial (that is, at their last known contact).

POISE-3 had a partial factorial design in which a subset of participants who were taking an antihypertensive medication also underwent randomization in the blood-pressure management trial [17]. We used Cox proportional hazards (time to the first event) models stratified by participation in the blood-pressure trial and by assigned blood-pressure strategy. We did not perform TXA $\times$ blood-pressure strategy interaction testing in this urology analysis because the POISE-3 trial found no evidence of effect modification for the primary efficacy or safety outcomes [15,17]. We reported hazard ratios (HRs) with two-sided 95% confidence intervals (CIs). For procedures (and approaches) with at least 50 patients, we also reported descriptive, procedure-specific estimates of the proportion of patients who experienced the primary efficacy and primary safety outcomes by the 30-day landmark in the TXA and placebo groups.

For subgroup analyses of the primary efficacy and safety outcomes, we used Cox proportional hazards models with tests of interaction. An interaction p value < 0.05 was taken to suggest the possible presence of a subgroup effect. Prespecified subgroups were surgical approach (open or minimally invasive vs transurethral; Supplement 1 provides details), active cancer at the time of surgery (yes vs no), baseline bleeding risk according to the 2026 International Society on Thrombosis and Haemostasis Scientific and Standardization Committee (ISTH SSC) classification [18], and antithrombotic use (therapeutic-dose thrombin or factor Xa inhibitor, therapeutic-dose vitamin K antagonist, therapeutic-dose intravenous or subcutaneous antithrombotic agent, prophylactic-dose anticoagulant, acetylsalicylic acid [ASA], and/or P2Y12 antagonists) in the 24 h preceding surgery.

For all analyses, we considered a two-sided p value < 0.05 to indicate statistical significance. We performed analyses using R software, version 4.2.3.

3. Results

3.1. Patients

After classifying procedures based on center responses, 1124 participants undergoing urologic surgery were included in this analysis (556 randomized to TXA and 568 to placebo, Supplement 4; 99% completed the 30-day follow-up; at least one postoperative troponin measurement in 509 [92%] in the TXA and 506 [89%] in the placebo group). Patients had a median age of 70 yr (interquartile range [IQR] = 65–75) and a median body mass index (BMI) of 28 (IQR = 25–32). A total of 208 patients (19%) had a history of myocardial infarction, 97 (8.6%) had a history of stroke, 100 (8.9%) had atrial fibrillation, and 658 (59%) had cancer. A total of 156 (14%) patients underwent robotic radical prostatectomy, 138 (12%) TURP, 86 (7.7%) laparoscopic radical nephrectomy, and 76 (6.8%) open radical cystectomy (Table 1). Surgical approach was laparoscopic/robotic in 489 (44%), open in 360 (32%), transurethral in 244 (22%), and percutaneous in 31 (2.8%). A total of 284 (25%) patients used antithrombotic medication in the 24 h preceding surgery. Table 1 summarizes patient characteristics, their relevance to primary efficacy and safety outcomes, and their distribution between the

TXA and placebo groups. In-hospital postoperative medication use was similar between arms (eg, thromboprophylaxis in 211/556 [38%] with TXA vs 227/568 [40%] with placebo; ASA in 108/556 [19%] vs 125/568 [22%]; and P2Y12 inhibitors in 7/556 [1.3%] vs 8/568 [1.4%], respectively).

3.2. Efficacy outcomes

By 30 days, the primary efficacy bleeding outcome occurred in 45 (8.1%) patients in the TXA group and in 62 (11%) patients in the placebo group (HR = 0.73, 95% CI = 0.50–1.17, Fig. 1, Table 2). Major bleeding occurred in 34 (6.1%) patients in the TXA group and in 54 (9.5%) in the placebo group (HR = 0.63, 95% CI = 0.41–0.97, Table 2). BIMS occurred in 42 (7.6%) patients in the TXA group and in 60 (11%) patients in the placebo group (HR = 0.70, 95% CI = 0.47–1.04, Table 3). Blood cell transfusions occurred in 48 patients (8.6%) in the TXA group and 64 (11%) in the placebo group (HR = 0.75, 95% CI = 0.52–1.10, Table 3). Tables 2 and 3 provide details of the efficacy outcomes.

3.3. Safety outcomes

By 30 days, the primary safety thrombosis outcome occurred in 67 (12%) patients in the TXA group and 62 (11%) patients in the placebo group (HR = 1.12, 95% CI = 0.79–1.58, Fig. 1, Table 2). MINS occurred in 62 (11%) patients in the TXA group and 58 (10%) patients in the placebo group (HR = 1.11, 95% CI = 0.78–1.59, Table 2). Nonhemorrhagic stroke occurred in 2 (0.4%) patients in each group (HR = 1.01, 95% CI = 0.14–7.15, Table 2), and symptomatic proximal VTE occurred in 3 (0.5%) patients in each group (HR = 1.03, 95% CI = 0.21–5.11, Table 2). Myocardial infarction occurred in 10 (1.8%) patients in the TXA group and in 4 (0.70%) patients in the placebo group (HR = 2.56, 95% CI = 0.80–8.17, Table 2). We identified no seizures in either group (Table 3). Tables 2 and 3 provide details of the safety outcomes.

3.4. Subgroup and procedure-specific analyses

We found no significant interaction for the primary efficacy bleeding or primary safety thrombosis outcomes by surgical approach (transurethral vs other), cancer status (present vs not present), or preoperative antithrombotic agent use (within 24 hours; yes vs no) (Table 4). We prespecified subgroup analyses by baseline bleeding risk using the 2026 ISTH SSC classification (high vs moderate/low/minimal). Since only 50 participants (4.4%) were moderate/low/minimal risk (vs 1074 [96%] high risk), we did not perform this analysis. Supplement 5 provides procedure-specific bleeding and safety outcome data (descriptive only, as the study was underpowered for procedure-specific comparative treatment effects).

4. Discussion

Our POISE-3 urology analysis is the largest randomized, placebo-controlled evaluation of TXA in urologic surgery. Among 1124 patients undergoing a broad range of urologic procedures and receiving two 1 g intravenous boluses of

Table 1 – Baseline characteristics by treatment assignment: tranexamic acid (TXA) or placebo

Variable	Included Entered study, n (%)	In this analysis by 30 d		Treatment assignment, n (%)	
		Primary efficacy (bleeding) outcome, n (%) ^a	Primary safety (thrombosis) outcome, n (%) ^a	TXA	Placebo
Number of patients	1124 (100)	107 (10)	129 (11)	556 (49)	568 (51)
Age (yr)					
Less than 50	18 (1.6)	1 (5.5)	1 (5.5)	11 (61)	7 (39)
50–69	656 (58)	66 (10)	83 (13)	335 (51)	321 (49)
70 or more	450 (40)	40 (8.9)	45 (10)	210 (47)	240 (53)
Sex					
Male	880 (78)	79 (9.0)	97 (11)	436 (50)	444 (50)
Female	244 (22)	28 (11)	32 (13)	120 (49)	124 (51)
Body-mass index (kg/m²)					
<25	284 (25)	34 (12)	35 (12)	133 (47)	151 (53)
25–30	444 (40)	40 (9.0)	50 (11)	228 (51)	216 (49)
30 or more	309 (27)	27 (8.7)	35 (11)	150 (49)	159 (51)
Missing	87 (7.7)	6 (6.9)	9 (10)	45 (53)	42 (48)
History of ...					
Hypertension	1019 (91)	102 (10)	117 (11)	504 (49)	515 (51)
Myocardial infarction	208 (19)	18 (8.7)	30 (14)	103 (50)	105 (50)
Stroke	97 (8.6)	9 (9.3)	14 (14)	44 (45)	53 (55)
Atrial fibrillation	100 (8.9)	12 (12)	18 (18)	51 (51)	49 (49)
Dementia	2 (0.17)	0	1 (50)	1 (50)	1 (50)
Active cancer	658 (59)	87 (13)	94 (14)	329 (50)	329 (50)
Receiving ongoing dialysis	1 (0.09)	1 (100)	0	0	1 (100)
Tobacco use	528 (47)	55 (10)	55 (10)	258 (49)	270 (51)
Coronary revascularization	239 (21)	19 (7.9)	38 (16)	116 (49)	123 (51)
PCI (bare-metal stent)	71 (6.3)	6 (8.5)	15 (21)	42 (59)	29 (41)
PCI (drug-eluting stent)	109 (10)	8 (7.3)	21 (19)	51 (47)	58 (53)
CABG	76 (6.8)	6 (7.9)	5 (6.6)	34 (45)	42 (55)
Geographical location					
North America	360 (32)	42 (12)	41 (11)	176 (49)	184 (51)
Europe	346 (31)	23 (6.6)	37 (11)	174 (50)	172 (50)
Africa-Middle East	4 (0.36)	0	1 (25)	1 (25)	3 (75)
Asia-Pacific	353 (31)	40 (11)	34 (10)	174 (49)	179 (51)
South America	61 (5.4)	2 (3.3)	16 (26)	31 (51)	30 (49)
Surgical approach					
Open	360 (32)	71 (20)	62 (17)	180 (50)	180 (50)
Laparoscopic/Robotic	489 (44)	30 (6.1)	54 (11)	244 (50)	245 (50)
Transurethral	244 (22)	3 (1.2)	4 (1.6)	119 (49)	125 (51)
Percutaneous	31 (2.8)	3 (10)	9 (29)	13 (42)	18 (58)
Surgical procedure					
Robotic radical prostatectomy	156 (14)	2 (1.3)	7 (4.5)	80 (51)	76 (49)
Laparoscopic radical nephrectomy	86 (7.7)	4 (4.7)	12 (14)	46 (53)	40 (47)
Laparoscopic partial nephrectomy	57 (5.1)	5 (8.8)	14 (25)	28 (49)	29 (51)
Robotic partial nephrectomy	56 (5.0)	2 (3.6)	4 (7.1)	30 (54)	26 (46)
Open radical nephrectomy	69 (6.1)	15 (22)	17 (25)	29 (42)	40 (58)
TURP/transurethral BPO surgery	138 (12)	1 (0.72)	2 (1.4)	74 (54)	64 (46)
Open radical cystectomy	76 (6.8)	30 (39)	14 (18)	41 (54)	35 (46)
Female pelvic floor surgery	64 (5.7)	0	8 (13)	33 (52)	31 (48)
TURBT	57 (5.1)	1 (1.8)	4 (7.0)	24 (42)	33 (58)
Other procedures	365 (32)	47 (13)	47 (13)	171 (47)	194 (53)
Medication taken within 24 h before surgery					
Therapeutic-dose thrombin or factor Xa inhibitor	6 (0.53)	1 (17)	1 (17)	2 (33)	4 (67)
Therapeutic-dose vitamin K antagonist	1 (0.09)	0	0	0	1 (100)
Therapeutic-dose intravenous or subcutaneous antithrombotic	9 (0.80)	4 (44)	2 (22)	4 (44)	5 (56)
Prophylactic dose anticoagulant	154 (14)	9 (5.8)	14 (9.1)	78 (51)	76 (49)
Acetylsalicylic acid (ASA)	129 (11)	14 (11)	17 (13)	54 (42)	75 (58)
P2Y12 antagonist	8 (0.71)	0	0	0	8 (100)
Nonsteroidal anti-inflammatory drug (NSAID)	20 (1.8)	4 (20)	2 (10)	6 (30)	14 (70)
Cyclooxygenase-2 (COX-2) inhibitor	18 (1.6)	2 (11)	0	5 (28)	13 (72)

BPO = benign prostatic obstruction; CABG = coronary artery bypass graft; PCI = percutaneous coronary intervention.

^a The primary efficacy outcome was a composite of life-threatening bleeding, major bleeding, or bleeding into a critical organ. The primary safety outcome was a composite of myocardial injury after noncardiac surgery, nonhemorrhagic stroke, peripheral arterial thrombosis, or symptomatic proximal venous thromboembolism. Outcome counts are descriptive (30-day events).

TXA or placebo, the primary efficacy bleeding outcome occurred in 8.1% in the TXA and 10.9% in the placebo group (HR = 0.73, 95% CI = 0.50–1.07). TXA reduced major bleeding

(6.1% vs 9.5%; HR = 0.63, 95% CI = 0.41–0.97). Life-threatening and critical-organ bleeding were uncommon. The reduction in major bleeding largely reflected fewer

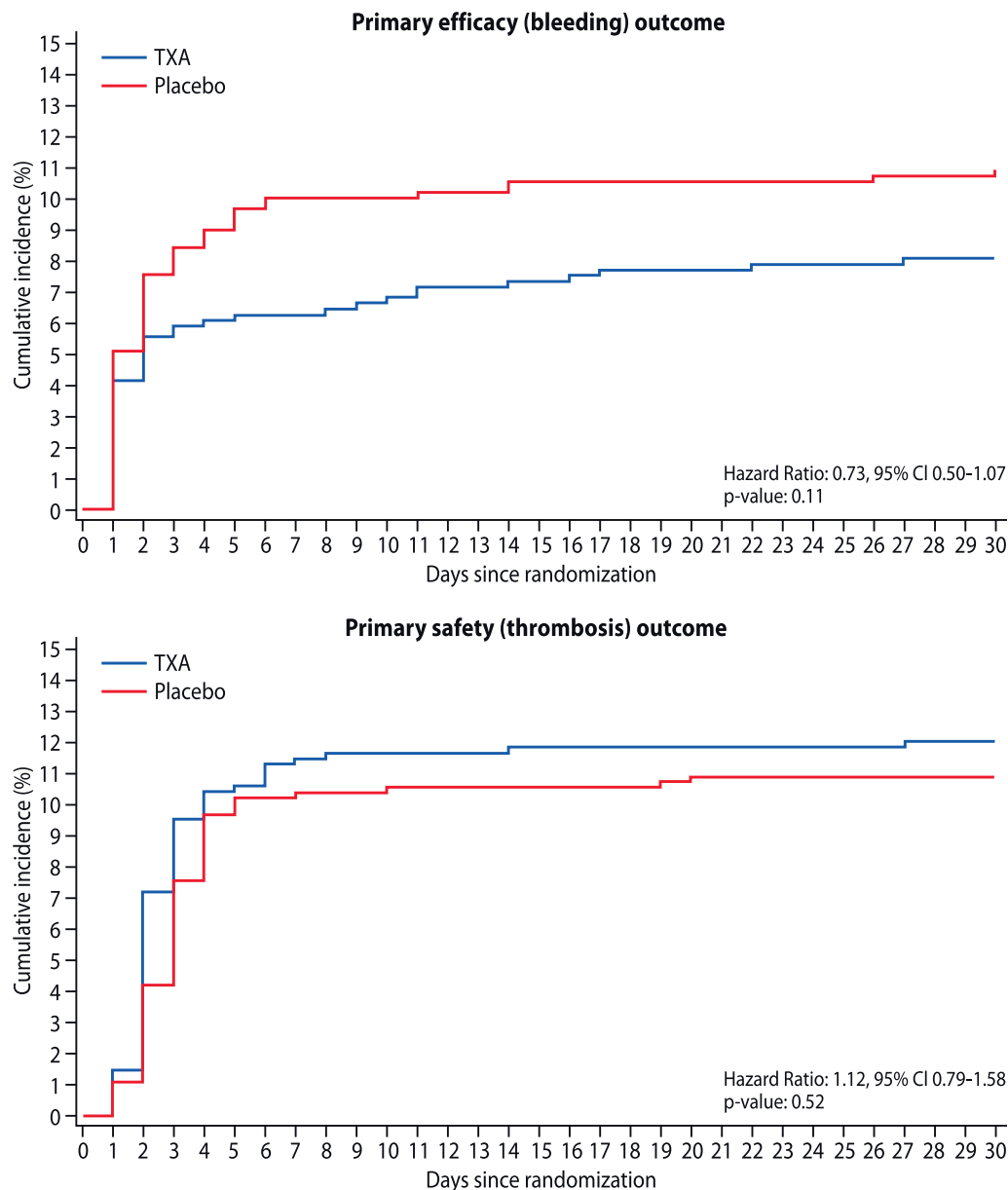


Fig. 1 – Kaplan-Meier estimates of the primary efficacy and safety outcomes.

transfusions. Fewer transfusions reduce exposure to blood products and their potential adverse effects and may also reduce health care resource use. Stroke (0.4% in both groups) and VTE (0.5% in both groups) were uncommon, and we found no clear increase in the primary safety thrombosis outcome. However, thrombosis estimates were imprecise, and CIs were wide. Overall, the absolute effects for an individual patient will depend on their baseline bleeding and thrombotic risk, which varies by procedure and patient characteristics.

Strengths of our study include the rigorous randomized, placebo-controlled, blinded design; international conduct across more than 100 hospitals; very high intervention compliance and completion of 30-day follow-up; and independent adjudication of bleeding and thrombosis outcomes. We finalized this statistical analysis plan before any analy-

ses and confirmed procedure types and approaches (open, minimally invasive, transurethral, and percutaneous) through structured chart review.

Importantly, the trial enrolled patients ≥ 45 yr with bleeding and cardiovascular risk factors [16], a population frequently encountered in perioperative care but often excluded from perioperative trials because comorbidities raise perceived safety concerns [19]. However, stroke (0.4%) and VTE (0.5%) rates were not high, and no fatal bleeding occurred. Therefore, our study population should not be viewed as uniformly high risk. The main limitation of our analysis is imprecision: despite being the largest randomized urology TXA evidence to date, power remained limited for several outcomes, particularly thrombosis events (and procedure-specific comparisons), resulting in wide CIs. Despite this limitation, the totality of data indi-

Table 2 – Efficacy and safety outcomes and their components by 30 days: Tranexamic acid (TXA) versus placebo

	TXA (n = 556)	Placebo (n = 568)	TXA vs placebo	
	n (%)	n (%)	HR (95% CI)	p value
Primary efficacy (bleeding) outcome	45 (8.1)	62 (11)	0.73 (0.50–1.07)	0.11
Life-threatening bleeding ^a	11 (2.0)	9 (1.6)	1.25 (0.52–3.02)	0.6
Major bleeding ^b	34 (6.1)	54 (10)	0.63 (0.41–0.97)	0.03
Critical organ bleeding	3 (0.54)	5 (0.88)	0.61 (0.15–2.55)	0.5
Primary safety (thrombosis) outcome	67 (12)	62 (11)	1.12 (0.79–1.58)	0.5
Myocardial injury after noncardiac surgery ^c	62 (11)	58 (10)	1.11 (0.78–1.59)	0.6
Non-hemorrhagic stroke	2 (0.36)	2 (0.35)	1.01 (0.14–7.15)	>0.9
Peripheral arterial thrombosis	0	0	-	-
Symptomatic proximal VTE	3 (0.54)	3 (0.53)	1.03 (0.21–5.11)	>0.9

VTE = venous thromboembolism.

P values and HR (95% CI) are from the stratified Cox proportional hazards regression models. P values and 95% CIs are two-sided. Supplement 1 (pages 18–25) provides outcome definitions.

^a Life-threatening bleeding was defined as bleeding that is fatal or leads to: significant hypotension that requires inotrope therapy, urgent (within 24 hours) surgery (other than superficial vascular repair), or intracranial hemorrhage.

^b Major bleeding was defined as bleeding that is not specified under “life-threatening bleeding” above, and results in a postoperative hemoglobin level ≤ 70 g/l; or the patient receives a transfusion of ≥ 1 unit of red blood cells; or leads to one of the following interventions (ie, embolization, superficial vascular repair, nasal packing).

^c Myocardial injury after noncardiac surgery was defined as any myocardial infarction (as defined below), and any elevated troponin (higher than the local lab threshold) judged to be due to myocardial ischemia (ie, without evidence of a non-ischemic etiology [eg, chronic elevation, pulmonary embolism, sepsis, cardioversion]) that occurred within the first 30 days after the initiation of surgery. The only exceptions to the definition of an elevated troponin will be to use a higher threshold for troponin T (TnT) of ≥ 30 ng/l, and for high-sensitivity troponin T (hsTnT) of 20 to <65 ng/l with an absolute change of at least 5 ng/l or an hsTnT level ≥ 65 ng/l.

Table 3 – Relative treatment effects of tranexamic acid (TXA) versus placebo on secondary and other tertiary outcomes by 30 days

Outcome	TXA n = 556 (%)	Placebo n = 568 (%)	HR (95% CI)
Secondary outcomes			
BIMS	42 (7.6)	60 (11)	0.70 (0.47–1.04)
Transfusions (proportion of patients) ^a	48 (8.6)	64 (11)	0.75 (0.52–1.10)
Transfusion of ≥ 2 units of packed red cells	37 (6.7)	43 (7.6)	0.88 (0.56–1.36)
Tertiary outcomes			
ISTH major bleeding	29 (5.2)	56 (9.9)	0.52 (0.33–0.81)
Hemorrhagic stroke	0	0	-
Myocardial infarction	10 (1.8)	4 (0.70)	2.56 (0.80–8.17)
Composite of peripheral arterial thrombosis and amputation	0	0	-
Symptomatic pulmonary embolism	3 (0.54)	2 (0.35)	1.55 (0.26–9.30)
Symptomatic proximal deep vein thrombosis	0	2 (0.35)	-
Seizures	0	0	-
Rehospitalization for vascular reasons	16 (2.9)	10 (1.8)	1.65 (0.75–3.64)
Death from any cause ^b	3 (0.54)	2 (0.35)	1.52 (0.25–9.10)

BIMS = Bleeding Independently associated with Mortality after noncardiac Surgery (defined as bleeding leading to a postoperative hemoglobin <70 g/l, requiring transfusion of red blood cells, and/or judged to be the immediate cause of death); ISTH major bleeding = International Society on Thrombosis and Hemostasis major bleeding (defined as a bleed that meets any of the following: i) fatal bleeding; ii) bleeding that is symptomatic and occurs in a critical area or organ, such as intracranial, intraspinal, intraocular, retroperitoneal, pericardial, in a non-operated joint, or intramuscular with compartment syndrome, assessed in consultation with the surgeon, and/or; iii) extrasurgical site bleeding causing a fall in hemoglobin level of 20 g/l (1.24 mmol/l) or more, or leading to transfusion of two or more units of whole blood or red cells, with temporal association within 24–48 h to the bleeding; iv) surgical site bleeding that requires a second intervention - open, arthroscopic, endovascular - or a hemarthrosis of sufficient size as to interfere with rehabilitation by delaying mobilization or delayed wound healing, resulting in prolonged hospitalization or a deep wound infection; or v) surgical site bleeding that is unexpected and prolonged and/ or sufficiently large to cause hemodynamic instability, as assessed by the surgeon. There should be an associate fall in hemoglobin level of 20 g/l (1.24 mmol/l), or transfusion, indicated by the bleeding, of at least 2 units of whole blood or red cells, with temporal association within 24 h to the bleeding).

^a Transfusions were not protocol mandated. Transfusion decisions were made by the treating clinicians according to local institutional policies and usual care.

^b Deaths in the TXA arm were due to hyperkalemia, cardiogenic shock and septic shock. In the placebo arm, deaths were due to septic shock and pulmonary embolism.

cates there is no clear increased risk of thrombosis. A large meta-analysis (191 randomized trials with >40 000 patients) found no clear difference in thrombotic events (relative risk = 1.02, 95% CI = 0.94–1.11, $p = 0.65$, $I^2 = 0$) [20].

Prior trials in TURP and PCNL have suggested that TXA can reduce perioperative bleeding and may reduce transfusion requirements, but overall certainty has been limited by methodological concerns [10,11,20]. In oncologic urology, a

randomized trial of 200 patients undergoing open radical retropubic prostatectomy (recruited 2008–2010) reported lower transfusion rates with TXA (34% vs 55%) [12]. However, the applicability of this evidence to contemporary practice may be somewhat reduced because radical prostatectomy is now performed predominantly robotically in many health systems, and transfusions are uncommon with current robotic techniques. In a recent Canadian random-

Table 4 – Relative treatment effects of tranexamic acid (TXA) versus placebo on the primary efficacy and safety outcomes by 30 days across subgroups

Outcome	TXA n/N (%)	Placebo n/N (%)	HR (95% CI)	Interaction p-value
Primary efficacy outcome: composite bleeding outcome				
Type of surgical approach				
Transurethral	1/119 (0.84)	2/125 (1.6)	0.59 (0.05-6.51)	0.8
Other*	44/437 (10)	60/443 (14)	0.73 (0.50-1.08)	
History of cancer				
Yes	35/329 (11)	52/329 (16)	0.66 (0.43-1.01)	0.4
No	10/227 (4.4)	10/239 (4.2)	1.04 (0.43-2.50)	
Patients with preoperative antithrombotic agents				
Yes	9/129 (7.0)	16/155 (10)	0.67 (0.30-1.52)	0.9
No	36/427 (8.4)	46/413 (11)	0.74 (0.48-1.15)	
Primary safety outcome: composite thrombosis outcome				
Type of surgical approach				
Transurethral	6/119 (5.0)	3/125 (2.4)	2.16 (0.54-8.69)	0.3
Other*	61/437 (14)	59/443 (13)	1.06 (0.74-1.51)	
Cancer				
Active cancer	52/329 (16)	42/329 (13)	1.27 (0.84-1.90)	0.2
No active cancer	15/227 (6.6)	20/239 (8.4)	0.79 (0.40-1.55)	
Patients with preoperative antithrombotic agents				
Yes	17/129 (13)	15/155 (9.7)	1.35 (0.67-2.70)	0.3
No	50/427 (12)	47/413 (11)	1.04 (0.70-1.56)	

* Other approaches include open, laparoscopic/robotic, and percutaneous.

ized trial of 353 adults undergoing open radical cystectomy, TXA did not reduce transfusion compared with placebo (37.0% vs 37.4%) [13]. In our analysis, although the reduction in transfusion with TXA did not reach statistical significance (8.6% with TXA vs 11% with placebo; HR = 0.75, 95% CI = 0.52–1.10), the point estimate was consistent with the observed reduction in major bleeding (HR = 0.63, 95% CI = 0.41–0.97). The treatment effect estimate on the primary efficacy bleeding outcome was less precise (HR = 0.73; 95% CI = 0.50–1.07), and the effects on life-threatening bleeding and critical-organ bleeding remained uncertain. However, both the overall effect and the effects on the components were consistent with the overall POISE-3 trial findings [15].

These findings can inform perioperative practice and policy. TXA is inexpensive and straightforward to administer, and preventing major bleeding is likely to avert clinically consequential downstream effects for patients and health systems. Patients generally prefer to avoid bleeding and transfusion [21]. However, available surveys suggest TXA remains infrequently used in routine urologic practice, including procedures with expected moderate blood loss, indicating potential underuse [6,8–10]. Given recurrent constraints in blood-product availability, wider adoption of TXA where appropriate could yield meaningful clinical and public-health benefits [22]. Guideline panels may therefore consider recommendations on perioperative TXA use in urology, particularly for patients at increased bleeding risk.

Important uncertainties remain. First, dosing regimens vary widely across trials. For longer, high-blood-loss operations such as radical cystectomy, it is plausible that dose, timing, or duration of antifibrinolytic exposure influences effectiveness [13,23]. This may also be modified by perioperative factors such as neoadjuvant chemotherapy. Second, the optimal route of administration for specific procedures is unclear: studies have used oral, intravenous, and local approaches (eg, irrigation for TURP or PCNL), but direct comparisons in urologic surgery are limited [10,11]. The

DEPOSITION trial provides strong evidence in cardiac surgery that topical TXA is inferior to intravenous TXA for red blood cell transfusion, with an absolute risk difference of 8.3% (95% CI = 5.2–11.5) [24]. Third, beyond antifibrinolysis, TXA may influence inflammatory or immune pathways via plasmin-related mechanisms [23]. Ongoing trials outside urology highlight interest in outcomes beyond bleeding [25,26].

5. Conclusions

Perioperative TXA reduced major (and ISTH major) bleeding in our prespecified POISE-3 urology analysis. We found no clear increase in thrombotic events. TXA is a reasonable option for urologic patients at increased bleeding risk, with decisions informed by procedure context, baseline risks, patient values, and local perioperative pathways.

Author contributions: Kari A. O. Tikkinen, Kumar Balasubramanian and P.J. Devereaux had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Tikkinen, Marcucci, Halme, Kanstrup, Kilpeläinen, Devereaux.

Acquisition of data: Marcucci, Ofori, Borges, Painter, Conen, Lomivorotov, Sessler, Chan, Martinez-Zapata, Wang, Xavier, Wang, Szczeklik, Leslie.

Analysis and interpretation of data: All authors.

Drafting of the manuscript: Tikkinen, Halme, Tondroanmag, Tornberg, Balasubramanian.

Critical revision of the manuscript for important intellectual content: All authors.

Statistical analysis: Balasubramanian.

Obtaining funding: Tikkinen, Marcucci, Devereaux.

Administrative, technical, or material support: Tikkinen, Devereaux.

Supervision: Tikkinen, Marcucci, Devereaux.

Financial disclosures: Kari A.O. Tikkinen and PJ Devereaux certify that all conflicts of interest, including specific financial interests and relationships and affiliations relevant to the subject matter or materials discussed in the manuscript (eg, employment/affiliation, grants or funding, consultancies, honoraria, stock ownership or options, expert testimony, royalties, or patents filed, received, or pending), are the following: Borges reported investigator-initiated grants from Roche International and Advisory board fees from Abbott. Conen reported advisory board fees from Abbott. Xavier reported grants from Boehringer Ingelheim, Bristol Myers Squibb, and a speaker fee from Eli Lilly. Devereaux reported grants from Abbott Diagnostics, AOP Pharma, AstraZeneca, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, CloudDX, Covidien, Javelin Medical Limited, Octapharma, Philips Healthcare, Roche Diagnostics, Siemens, and Stryker, and consultant fees from AstraZeneca, Roche Canada, Rafa Laboratories Ltd, and Trimedic. All other authors declare no competing interests.

Funding/Support and role of the sponsor: Open access was funded by the Helsinki University Library. Funding for this study came from several grants for the POISE-3 trial and its substudies: Canada: Foundation Grant (FDN-143302) from the Canadian Institutes of Health Research; Australia: Project Grant (1162362) from the Australian National Health and Medical Research Council; China: Grant from General Research Fund 14104419, Research Grant Council, Hong Kong SAR, and Population Health Research Institute. Finland: Competitive Research Funding of the Helsinki University Hospital (TYH2025385) and Sigrid Jusélius Foundation. Flavia K. Borges and David Conen hold PHRI Career Awards. The sponsors had no role in the study design or in manuscript preparation, review or approval. No third party was involved in the study design, conduct, data collection, analysis, interpretation or manuscript preparation.

Acknowledgments: We thank the patients, project office staff, especially Ingrid Copland and Louise Mastrangelo, and health care personnel involved in the study.

Data sharing: We believe that the dissemination of clinical research results is vital and that data sharing is important. We prioritize access to data analyses for researchers who have worked on the trial for a significant period, have played substantial roles, and have participated in raising the funds to conduct the trial. This balances the duration of the research study and the intellectual and financial investments that made it possible with the need to allow wider access to the data collected. Data will be disclosed only upon request and approval of the proposed use of the data by a review committee. Data requests from other non-POISE-3 Trial investigators will not be considered until March 2028. All data requests should be made to the last author (PJ.Devereaux@phri.ca).

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eururo.2026.03.019>.

References

- [1] Tikkinen KAO, Craigie S, Agarwal A, et al. Procedure-specific risks of thrombosis and bleeding in urological non-cancer surgery: systematic review and meta-analysis. *Eur Urol* 2018;73:236–41.
- [2] Tikkinen KAO, Craigie S, Agarwal A, et al. Procedure-specific risks of thrombosis and bleeding in urological cancer surgery: systematic review and meta-analysis. *Eur Urol* 2018;73:242–51.
- [3] Vascular Events in Noncardiac Surgery Patients Cohort Evaluation (VISION) Study Investigators, Spence J, LeManach Y, et al. Association between complications and death within 30 days after noncardiac surgery. *CMAJ* 2019;191:E830–7.
- [4] Dunn CJ, Goa KL. Tranexamic acid: a review of its use in surgery and other indications. *Drugs* 1999;57:1005–32.
- [5] Poeran J, Chan JJ, Zubizarreta N, Mazumdar M, Galatz LM, Moucha CS. Safety of tranexamic acid in hip and knee arthroplasty in high-risk patients. *Anesthesiology* 2021;135:57–68.
- [6] Warnakulasuriya SR, Wolff K, Stanworth SJ, of the Research and Audit Federation of Trainees group and the NIHR BTRU in Data Driven Transfusion Practice Collaborators. Provision of peri-operative patient blood management strategies in the UK: a national survey of practice. *Anaesthesia* 2025;80:781–9.
- [7] Painter TW, McLroy D, Myles PS, Leslie K. A survey of anaesthetists' use of tranexamic acid in noncardiac surgery. *Anaesth Intensive Care* 2019;47:76–84.
- [8] 2024 National comparative audit of NICE Quality Standard QS138. Hospitals and Science – NHSBT. 2024. Accessed February 15, 2026. <https://hospital.blood.co.uk/audits/national-comparative-audit/reports-grouped-by-year/2024-national-comparative-audit-of-nice-quality-standard-qs138/>.
- [9] Montroy J, Hutton B, Fergusson DA, et al. Lysine analogue use during cancer surgery: a survey from a Canadian tertiary care centre. *Curr Oncol* 2020;27:e560–8.
- [10] Oliveira AV, Sampaio ALN, Freitas GA, et al. Tranexamic acid in men undergoing transurethral resection of prostate: an updated systematic review and meta-analysis. *Prostate* 2026;86:279–90.
- [11] Cleveland B, Norling B, Wang H, et al. Tranexamic acid for percutaneous nephrolithotomy: an abridged Cochrane review. *BJU Int* 2024;133:259–72.
- [12] Crescenti A, Borghi G, Bignami E, et al. Intraoperative use of tranexamic acid to reduce transfusion rate in patients undergoing radical retropubic prostatectomy: double blind, randomised, placebo controlled trial. *BMJ* 2011;343:d5701.
- [13] Breau RH, Lavallée LT, Cagiannos I, et al. Tranexamic acid during radical cystectomy: a randomized clinical trial. *JAMA Surg* 2024;159:1355–63.
- [14] Mangold MH, Wieland VLS, Grilli M, et al. Tranexamic acid in radical cystectomy: a systematic review and meta-analysis of efficacy and safety. *World J Urol* 2025;43:560.
- [15] Devereaux PJ, Marcucci M, Painter TW, et al. Tranexamic acid in patients undergoing noncardiac surgery. *N Engl J Med* 2022;386:1986–97.
- [16] Marcucci M, Painter TW, Conen D, et al. Rationale and design of the PeriOperative ischemic Evaluation-3 (POISE-3): a randomized controlled trial evaluating tranexamic acid and a strategy to minimize hypotension in noncardiac surgery. *Trials* 2022;23:101.
- [17] Marcucci M, Painter TW, Conen D, et al. Hypotension-avoidance versus hypertension-avoidance strategies in noncardiac surgery: an international randomized controlled trial. *Ann Intern Med* 2023;176:605–14.
- [18] Shaw JR, Abdulrehman J, Stella SF, et al. Surgical and procedural bleed risk stratification for anticoagulated patients undergoing planned surgery: guidance from the ISTH SSC subcommittee on perioperative and critical care thrombosis and hemostasis. *J Thromb Haemost*. In press. <https://doi.org/10.1016/j.jth.2025.12.031>.
- [19] Yates J, Perelman I, Khair S, et al. Exclusion criteria and adverse events in perioperative trials of tranexamic acid: a systematic review and meta-analysis. *Transfusion* 2019;59:806–24.
- [20] Tsan SEH, Viknaswaran NL, Cheong CC, Cheah S, Ng KT, Mong SXY, et al. Prophylactic intravenous tranexamic acid and thromboembolism in non-cardiac surgery: a systematic review, meta-analysis and trial sequential analysis. *Anaesthesia* 2023;78(9):1153–61.
- [21] Vetter TR, Adhami LF, Porterfield JR, Marques MB. Perceptions about blood transfusion: a survey of surgical patients and their anesthesiologists and surgeons. *Anesth Analg* 2014;118:1301–8.
- [22] Roberts I, Murphy MF, Moonesinghe R, et al. Wider use of tranexamic acid to reduce surgical bleeding could benefit patients and health systems. *BMJ* 2024;385:e079444.
- [23] Lam T, Medcalf RL, Cloud GC, Myles PS, Keragala CB. Tranexamic acid for haemostasis and beyond: does dose matter? *Thromb J* 2023;21:94.
- [24] Lamy A, Sirota DA, Jacques F, et al. Topical versus intravenous tranexamic acid in patients undergoing cardiac surgery: the DEPOSITION randomized controlled trial. *Circulation* 2024;150:1315–23.
- [25] Australian and New Zealand College Of Anaesthetists, TRIGS trial: tranexamic acid to Reduce Infection after gastrointestinal Surgery. Accessed February 15, 2026. <https://www.anzca.edu.au/research/>

[anzca-clinical-trials-network/our-clinical-trials/clinical-trials-pending-publication-and-announcement-of-results/trigs-trial-tranexamic-acid-to-reduce-infection-after-gastrointestinal-surgery](#).

[26] Australian New Zealand Clinical Trials Registry. A feasibility trial of tranexamic acid for necrotising soft-tissue infections. Accessed February 15, 2026. <https://anzctr.org.au/Trial/Registration/TrialReview.aspx?id=382739&isReview=true>.